

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Physics

2. **Subject Name:** Atomic and Molecular Physics **L-T-P:** 3-0-0

Credits: 3

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus:

Atomic structure: Spontaneous emission, Line spectra, Angular momentum and Spin, Addition of angular momenta and C.G. coefficients, Spin-Orbit coupling and Fine structure of spectral lines, Spectra of alkali atoms, vector atom model, Multiplet structure., Selection rules, R-S coupling and j-j coupling schemes, Normal and Anomalous Zeeman effect, Polarisation, Stark effect, Nuclear spin and hyperfine structure. (20)

Symmetric and antisymmetric wave functions, Slater determinants, constant field approximation, Hartree-Fock method, Born-Oppenheimer approximation. Rotational and Vibrational spectra. Spectra of diatomic molecules, polyatomic molecules. (14)

Raman Spectroscopy, magnetic resonance, ESR and NMR spectra, interaction of laser with atoms. (8)

Reference Books:

1. Physics of Atoms and Molecules: B.H. Bransden, C.J. Joachain
2. Fundamentals of Molecular & Spectroscopy. C. N. Banwell.
3. Atomic spectra and atomic structure, G. Herzberg

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Atomic structure: Spontaneous emission, Line spectra, Angular momentum and Spin, Addition of angular momenta and C.G. coefficients, Spin-Orbit coupling and Fine structure of spectral lines, Spectra of alkali atoms, vector atom model, Multiplet structure., Selection rules, R-S coupling and j-j coupling schemes, Normal and Anomalous Zeeman effect, Polarisation, Stark effect, Nuclear spin and hyperfine structure. (20)	20
2.	Symmetric and antisymmetric wave functions, Slater determinants, constant field approximation, Hartree-Fock method, Born-Oppenheimer approximation. Rotational and Vibrational spectra. Spectra of diatomic molecules, polyatomic molecules. (14)	14
3.	Raman Spectroscopy, magnetic resonance, ESR and NMR spectra, interaction of laser with atoms. (8)	8
Total number of hours		42

